Data Storytelling

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Spring 2026

Week 07

Announcements

- Next week: Spring Break! No class on Monday.
- Updated syllabus
- Final part of class: find a group and brainstorm a project

Project Proposal

- Should include 2-3 people per group
- Due: **March 30th at 11:59 PM**
- ~5 pages, including:
  - *Introduction* (Who is your audience? What is the question you are trying to answer?)
  - *Motivation/Theory* (Why is it interesting? What literature is this related to?)
  - *Data* (What data will you use? How will you get it?)
  - *Methods* (Will this be data exploration? Statistical analysis? Prediction?)
  - *Expected Results* (What do you expect to find? What would be interesting?)
- References do not count towards the page limit
- Use Google Docs or Overleaf and share pdf + link to Google Docs on Canvas

Project Proposal Suggestions

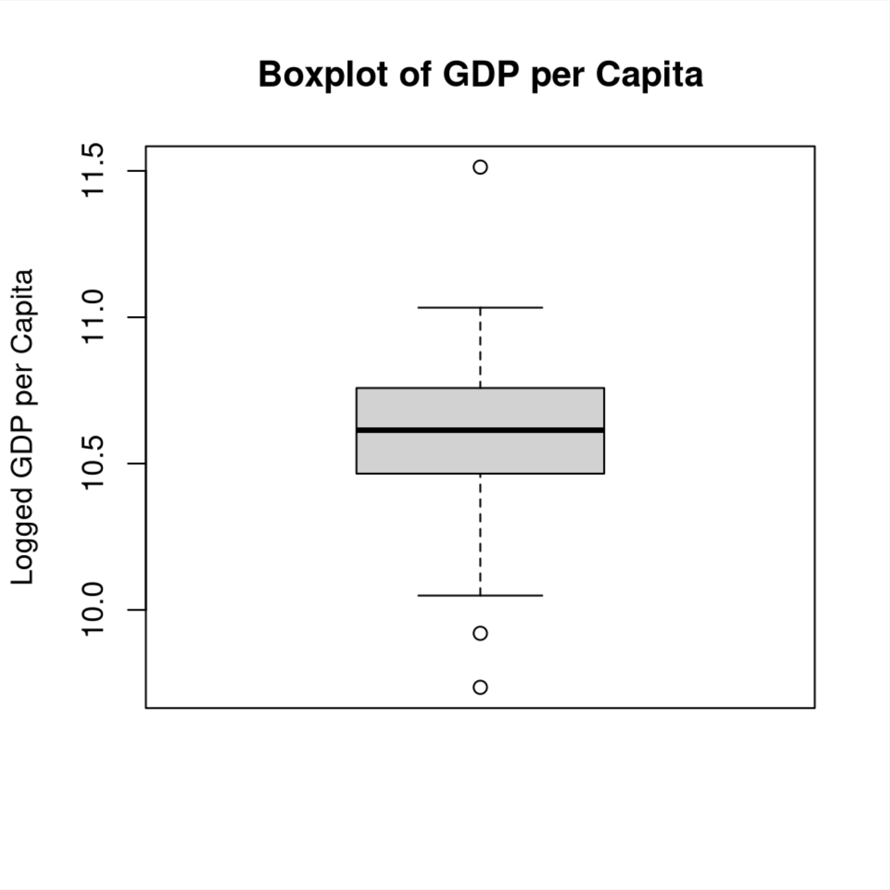
- **Project types:**
  - *Data exploration:* What interesting patterns can you find in a dataset?
  - *Statistical analysis:* Can you test a hypothesis using data?
  - *Prediction:* Can you build a model to predict an outcome of interest?
- **Audiences:**
  - *News consumer:* if you are exploring some phenomenon, you can write your paper as a piece of journalism that would be interesting to a general audience
  - *Academic:* if you are testing a hypothesis or describing a relationship between variables, you can speak to a theoretical literature
  - *Policy makers:* policy makers are often interested in understanding the effect or consequences of policies

Pitfalls of Data Analysis

Outliers

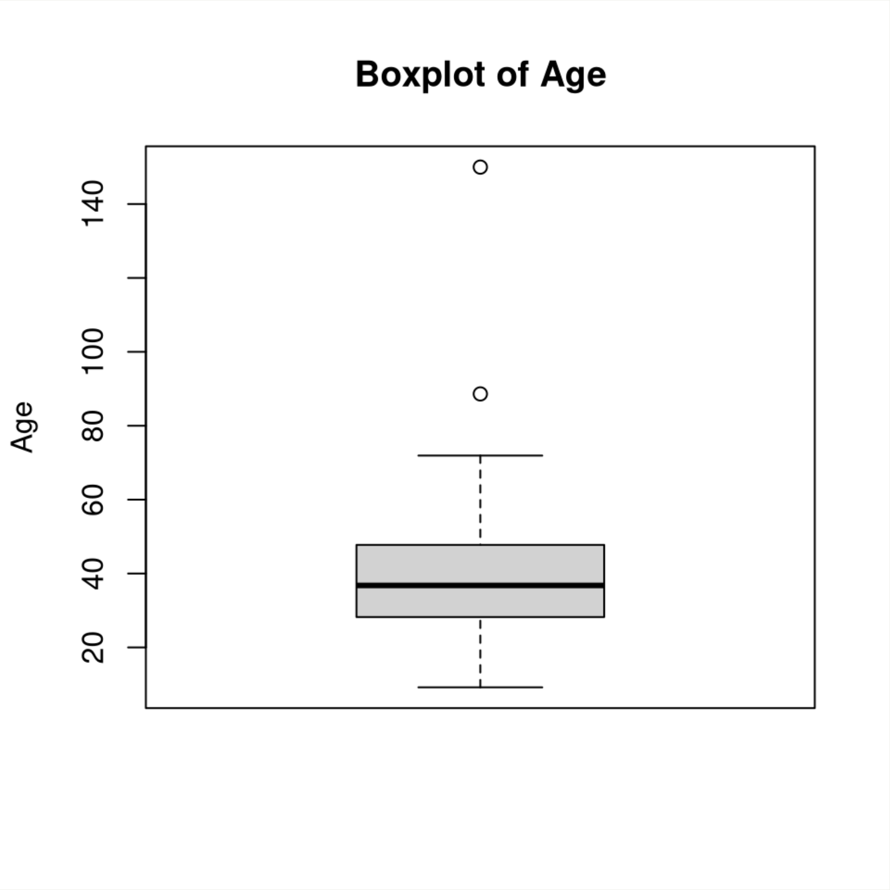
- Outliers are data points that are significantly different from the rest of the data. They can occur due to measurement errors, data entry errors, or they may be genuine observations that are rare or extreme.
  - If the outlier is due to a data entry or measurement error, it should be removed from the dataset to prevent it from skewing the analysis.
  - If the outlier is a byproduct of the data production process, it should be retained in the dataset, as it may provide valuable information about the underlying phenomenon being studied.

Outliers: Example 1



- Does this distribution look plausible? How about the outlier?
- How would you handle it?

Outliers: Example 2



- Does this distribution look plausible? How about the outlier?
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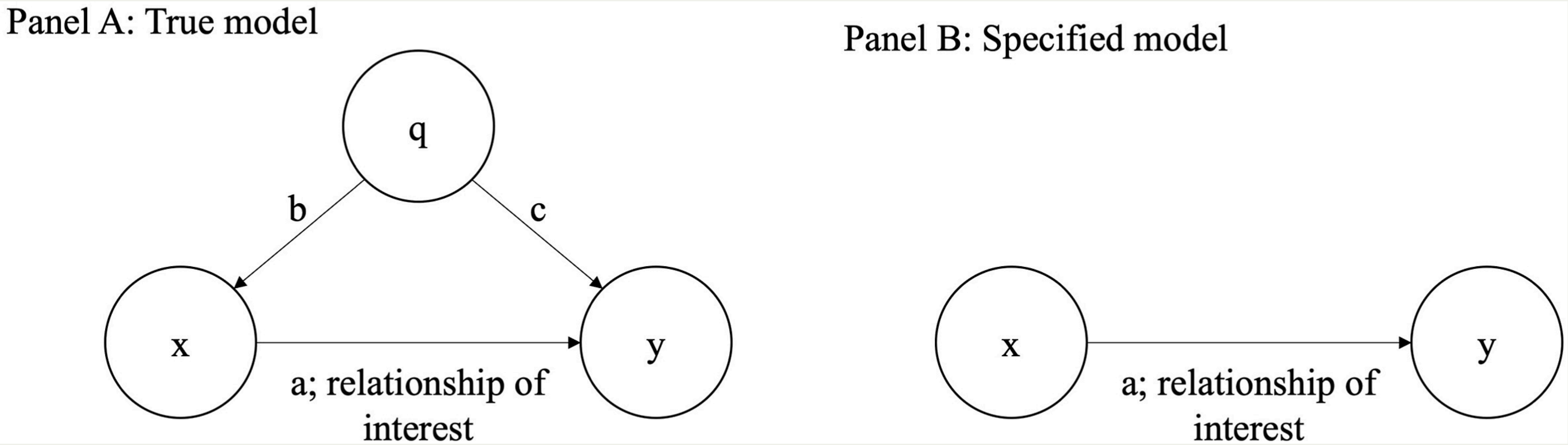
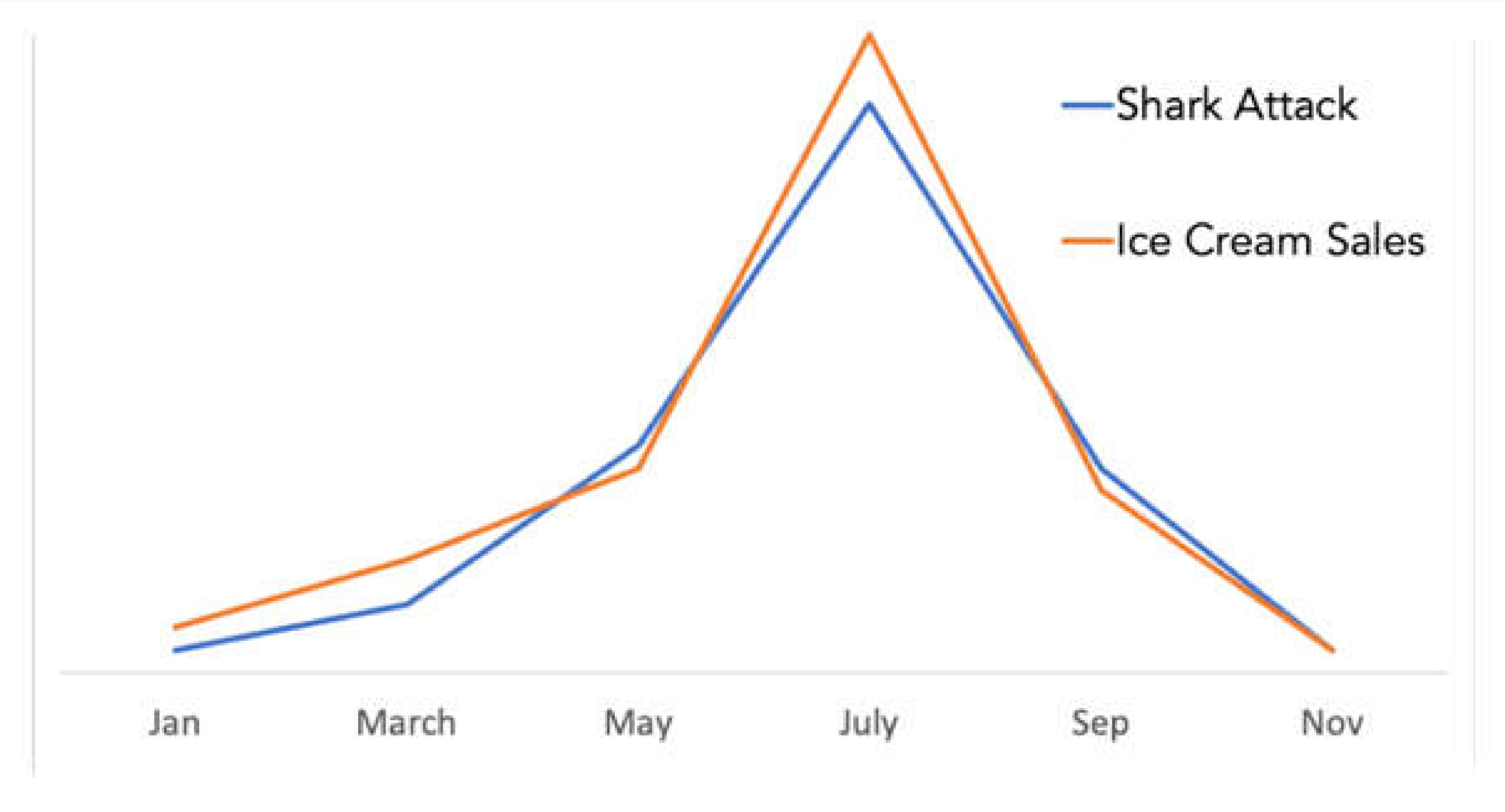
Omitted Variable Bias



- Classic correlation != causation
- How do you know? Ideally, you would control for all other variables that could be driving the relationship.
    - Ideally, use an experiment

Omitted Variable Bias



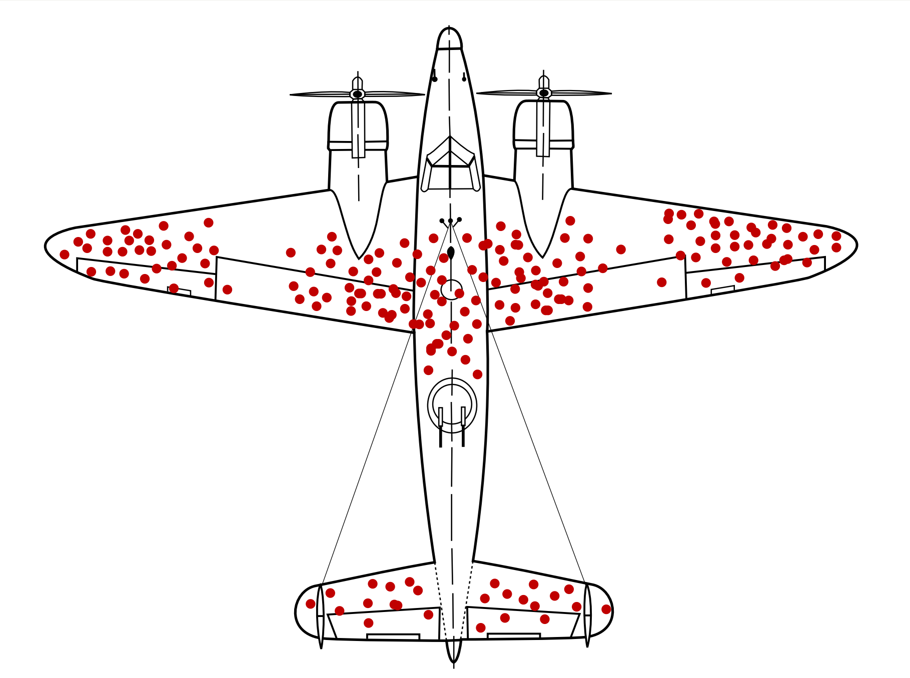
- Association between shark attacks and ice cream sales.

Survivorship Bias

> "My grandmother smoked 2 packs of cigarettes a day and lived to be 90, so smoking can't be that bad for you."

- A form of *selection bias* when we focus on the successful cases or "survivors" of a process while ignoring the unsuccessful cases or "non-survivors." This can lead to a distorted view of reality and incorrect conclusions.
  - "If you want to be a successful entrepreneur, just look at the stories of successful entrepreneurs and copy what they did!" (we don't see the ones that failed)
  - "Romans sure built strong buildings, look at the roads that are still around today!" (we don't see the ones that collapsed)

Plane Example of Survivorship Bias



- “Clearly, we need to add more armor to the wing tips and tail of the plane, since those are the areas that are getting hit the most!” (we don’t see the planes that got hit in the body and didn’t make it back)

Data Visualization

Cognitive Load

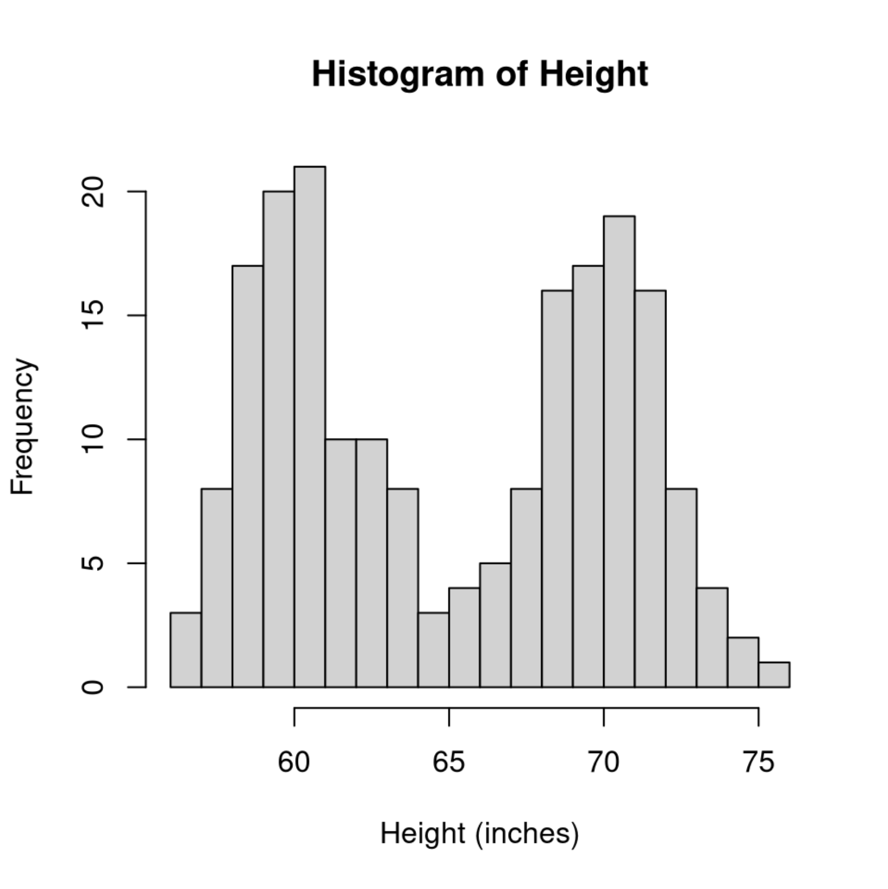
- It's more than just adding colors and labels.
- Our brains have finite attention and memory.
- Presenting data in intuitive ways can reduce *cognitive load* and make it easier for our brains to process the information.
- What contributes to *cognative load*? Clutter
  - Visual elements that take up space but don't add information

Gestalt principles of visual perception

- Gestalt principles are a set of rules that describe how our brains organize visual information.
  - *Proximity:* Objects that are close together are perceived as a group.
  - *Similarity:* Objects that are similar in shape, color, or size are perceived as a group.
  - *Continuity:* Our brains prefer to see continuous lines and patterns rather than disjointed ones.
  - *Closure:* Our brains tend to fill in missing information to create a complete image.
  - *Enclosure:* Objects that are enclosed within a boundary are perceived as a group.
  - *Connection:* Objects that are connected by lines or other visual elements are perceived as a group.

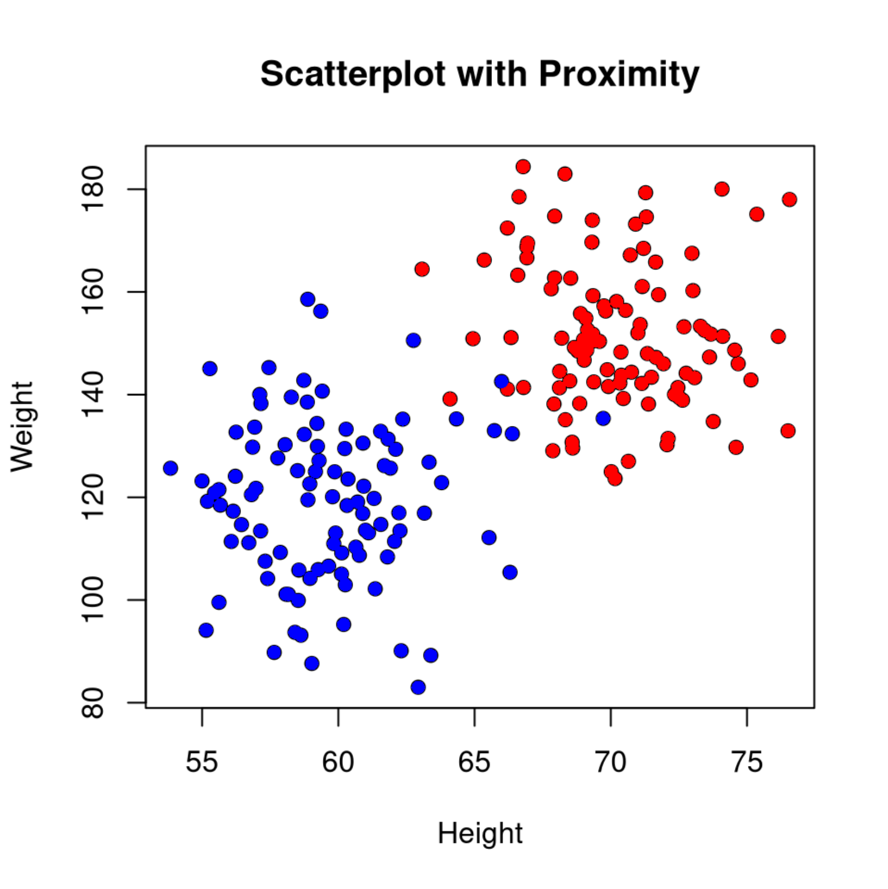
Proximity

- We tend to think of objects that are physically close together as belonging to part of a group
  - The logic underlying clustering algorithms
- Ex. If you create a histogram of height, you'll tend to see bimodal distribution if you include both men and women



Similarity

- Objects that are of the same color, shape, size, or orientation are perceived as belonging to the same group
  - Ex. If you create a scatterplot of height and weight, you'll tend to see two clusters if you color the points
- Can leverage this in tables to highlight rows you want to draw attention to

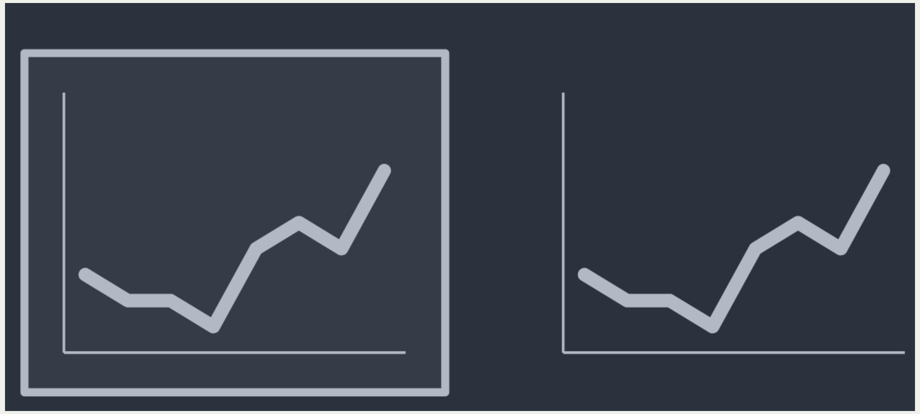


Enclosure

- Objects that are enclosed within a boundary are perceived as belonging to the same group
  - I.e. Shade in a region in a forecast model to highlight the predicted vs actual values
- *Example*
  - If you are forecasting future job growth, you can shade projected job growth in a different color to highlight the forecasted values

Closure

- The closure concept says that people like things to be simple and to fit in the constructs that are already in our heads.
  - People tend to perceive a set of individual elements as a single, recognizable shape when they can—when parts of a whole are missing, our eyes fill in the gap

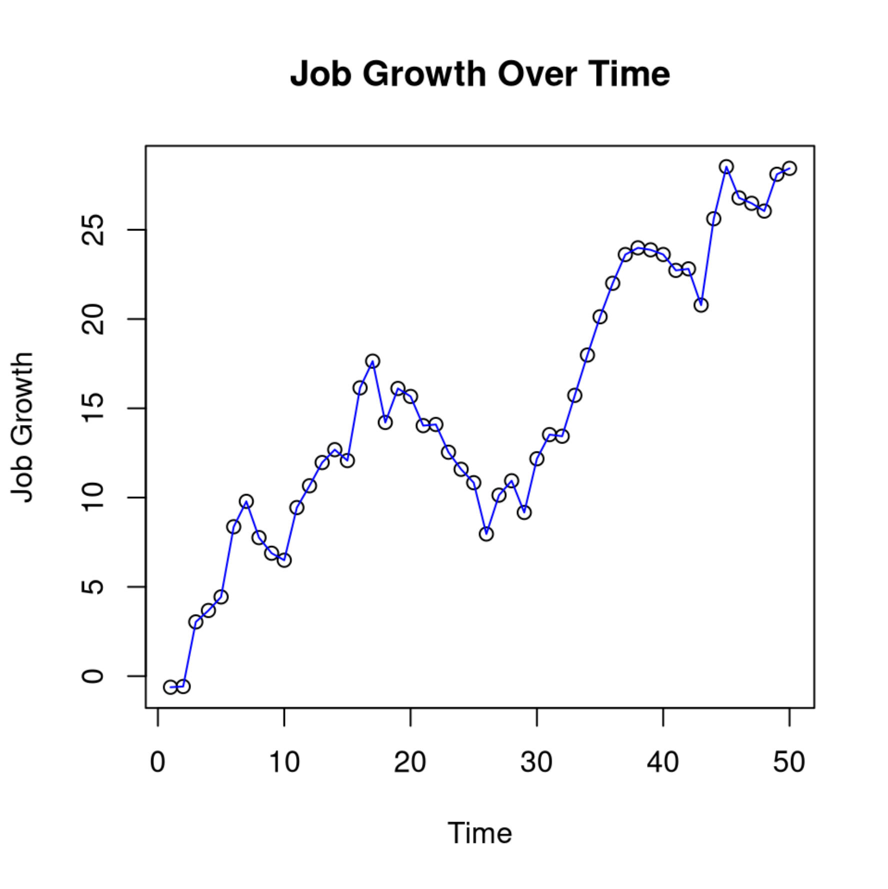


Continuity

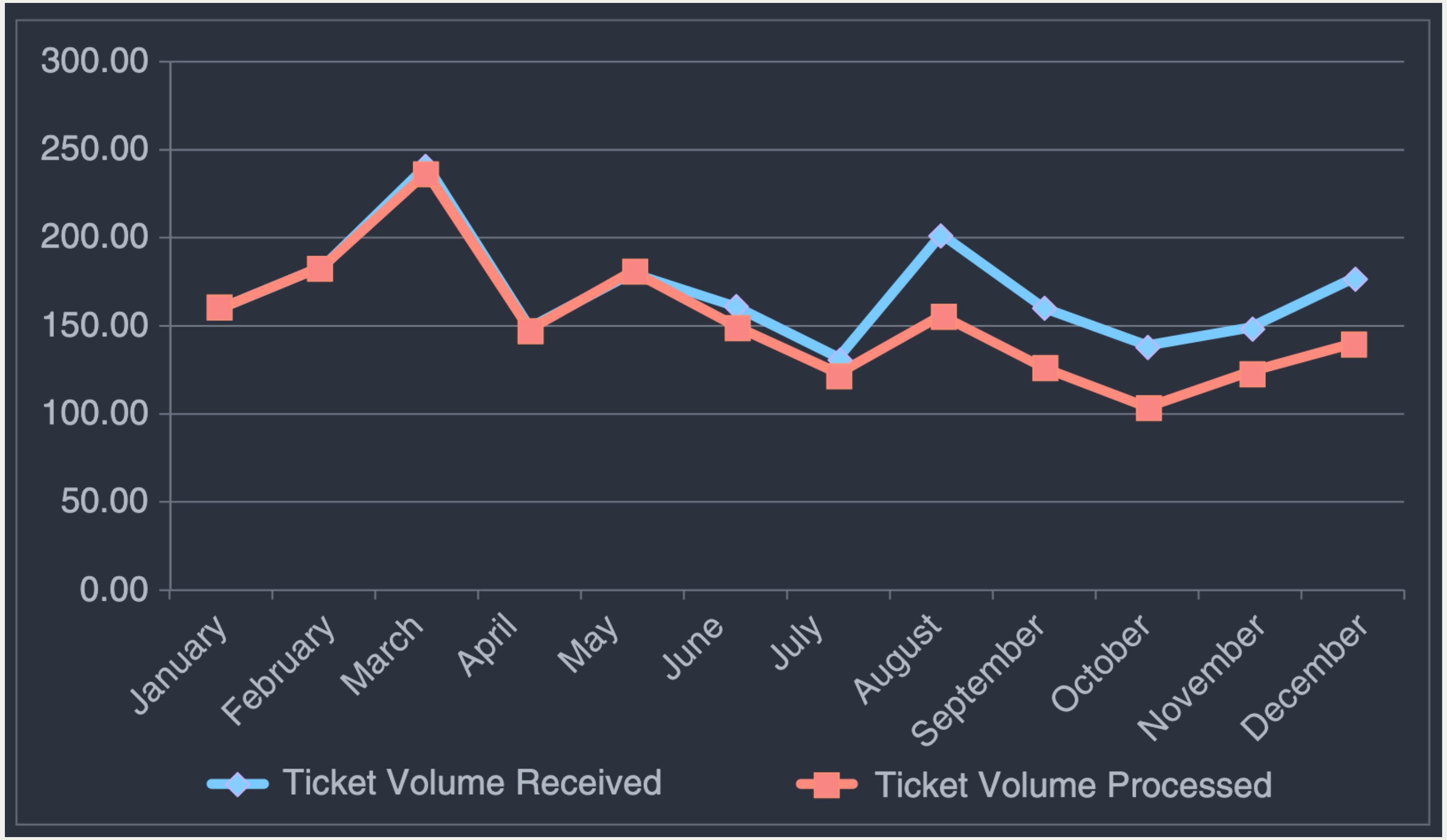
- When looking at objects, our eyes seek the smoothest path and naturally create continuity in what we see even where it may not explicitly exist
  - Similar to closure, but more about how we perceive lines and patterns rather than shapes

Connection

- We tend to think of objects that are physically connected as part of a group.
  - Ex. If we examine the trend of job growth over time, we will tend to see a pattern in the data if we connect the points with a line



Examples: Ticket Processing



- How would you improve upon this graph?

Examples: Ticket Processing Improvement